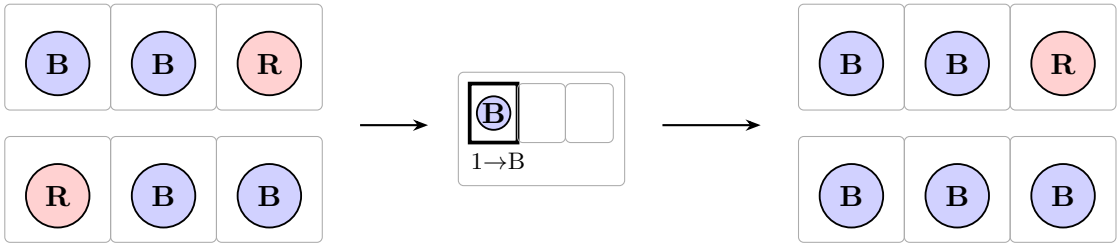
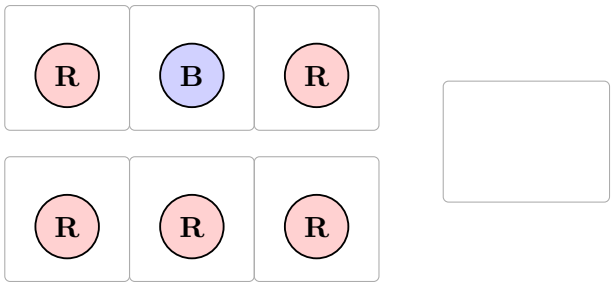
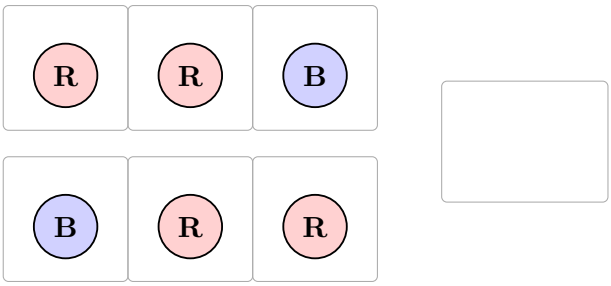
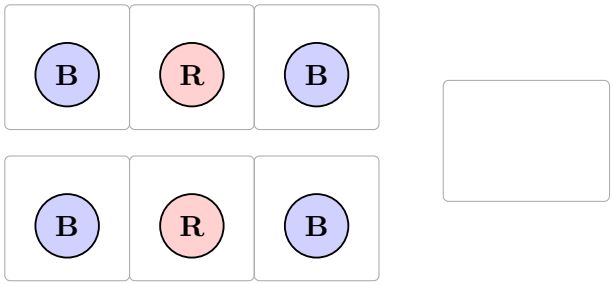
Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.

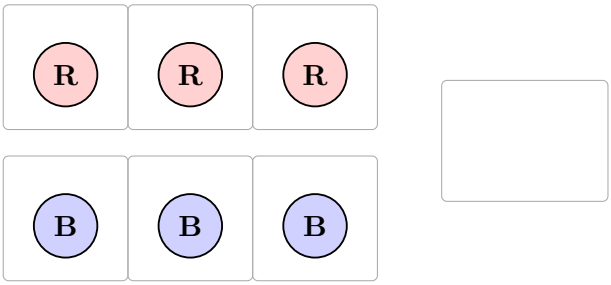


Problem 1: How does the fewest number of instructions needed for a match depend on the starting pair? Develop a rule that works for every pair.

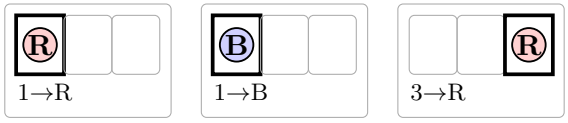


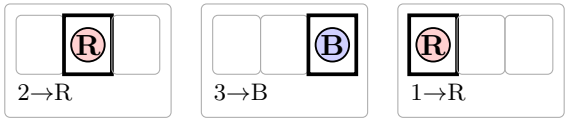


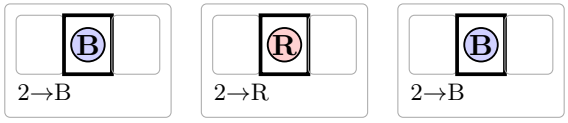




Problem 2: For each instruction story, how many different final boards can come from the eight possible starts? Determine exactly which starting colors still matter.

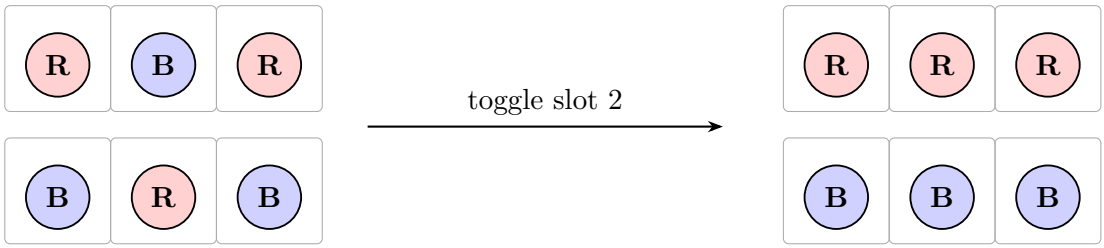




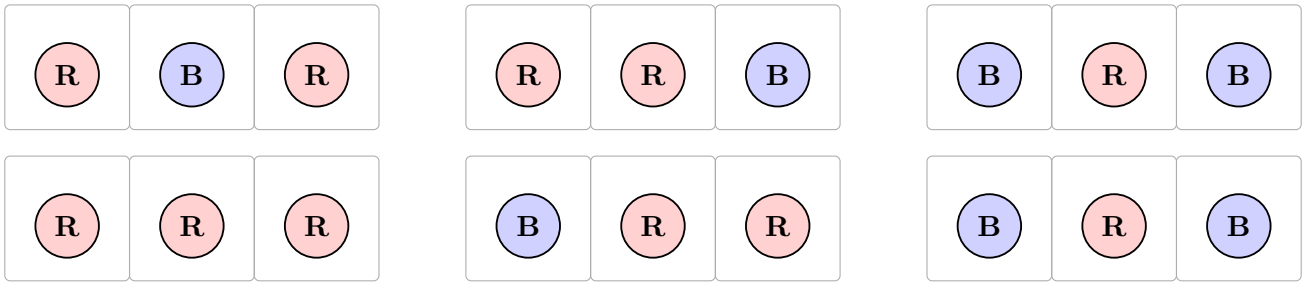


Problem 3: Explain exactly when an instruction story forgets every starting color. Could two different starts agree before that happens?

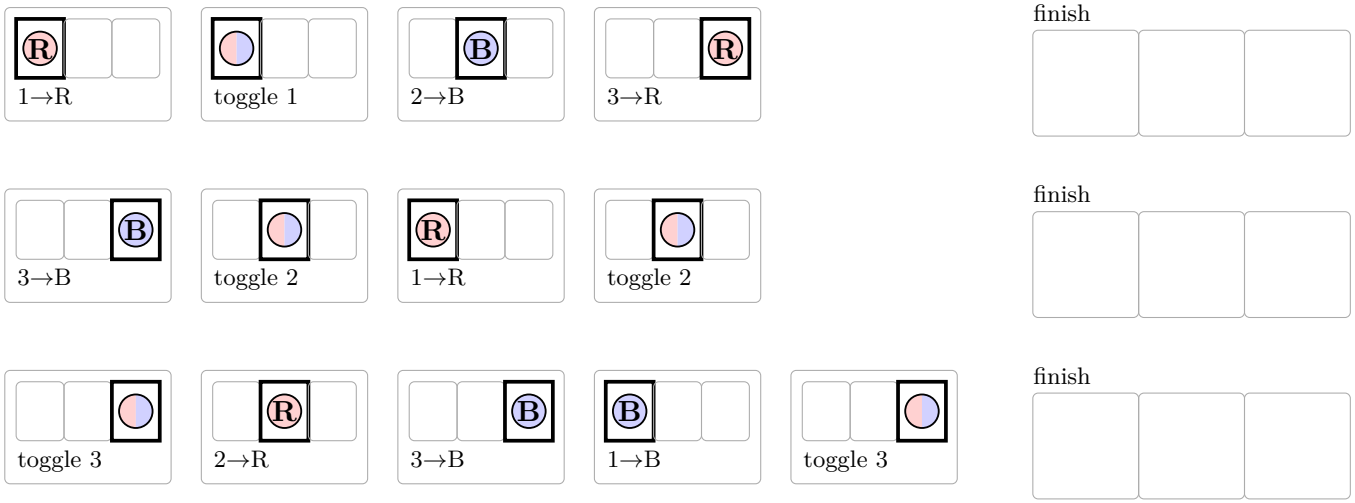
A toggle changes red to blue or blue to red in its slot, on both boards. It never sets a chosen color.



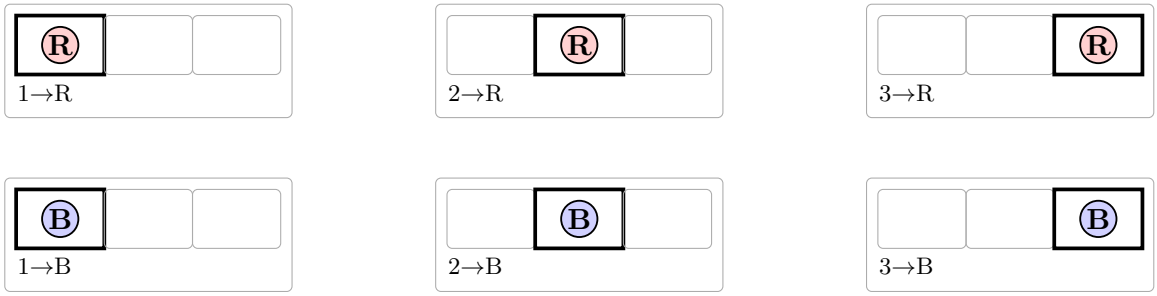
Problem 4: Can shared toggles erase a disagreement between two starting boards? Give an argument for every possible sequence of toggles.



Problem 5: Allow both color-setting instructions and toggles, as in these stories. Find a condition on the whole instruction story that guarantees every start gives the same finish. Explain why it works and why every story without it fails.



Draw independently from six equally likely cards, replacing each one.



Problem 6: Suppose four draws name slots 1, 3, 1, 2, in that order. Among the 16 equally likely color stories, which final boards have the greatest chance? Explain without assuming any starting colors.

Problem 7: Would your argument work for any slot story that visits all three slots? Does it mean four random instructions must forget the start?